

Big Data Analytics with Spark SQL

Big Data Analysis

Introduction to Spark SQL

Apache Spark SQL is a Spark module that uses the *DataFrame* abstraction to simplify working with structured data. Spark *DataFrames* provide a layer of abstraction on top of *RDDs*. It is effectively a collection of data that is organized into columns similar to a table in a relational store.

Data is represented in a Spark data frame as follows: a Spark data frame contains data in a *column* and *row* format where each column is a feature or variable and each row is a record. Spark data frames are able to read data from a multitude of data sources. Once the data is within a data frame, several operations (i.e., transformations) can be performed on the data. After one or more transformations are applied, the data is either output/displayed or can be leveraged for further data processing. Spark data frames can be created from existing *RDDs*, structured data files, external databases, or tables in Hive.

NOTE: *This document contains captured images from this module's Colab exercises. To view data contained in these images in a format amenable to screen readers, please refer to <https://colab.research.google.com/drive/1rZrDvXD0NGexmGGOmKa9jtSZ3DJwzKlX#scrollTo=2qY9HjthtVDs>*

Key Characteristics of Spark SQL

- **Integration with Spark:** Thanks to Spark SQL, relational data processing is integrated with Spark's functional/procedural programming approach. Spark SQL allows you to query structured data from within a Spark program using SQL in conjunction with the *DataFrame* API.
- **Hive compatibility:** Spark SQL, which is fully compatible with Hive queries and user-defined functions (UDFs), allows Hive queries to be executed unmodified on existing Hive data Warehouses. Furthermore, the metadata of Hive tables can be accessed via Spark SQL.
- **Unified data access:** Spark SQL allows data to be loaded and queried from numerous data sources including Hive tables, Parquet files, JSON files, Apache ORC, and Avro. In addition, data from across these disparate data sources can be joined together.
- **Standard connectivity:** The connectivity to Business Intelligence tools is via standard connectivity protocols such as ODBC and JDBC.

- **Performance and scalability:** Spark SQL comprises a cost-based optimizer, column-based storage, and dynamic code generation resulting in extremely agile query execution. Furthermore, Spark can scale across multiple nodes and enable multi-hour queries using the Spark engine, which supports full mid-query fault tolerance to facilitate recoverability in the event of failures.
- **Community:** Because Spark SQL is included with Apache Spark, it is tested and updated with each new Spark release. There is a thriving community of developers who contribute toward the development and testing of Spark and its associated libraries and APIs.

Spark SQL Operations: An Overview

Before any queries can be performed on data, the underlying dataset needs to be read into a data frame. However, before reading the data in, a connection needs to be established with Spark using the *SparkSession* module/package. Below are the steps to accomplish this.

Step 1

Establish connectivity with Spark using the *SparkSession* module/package.

```
# Import SparkSession from pyspark.sql
from pyspark.sql import SparkSession

# Create spark_session
spark_session = SparkSession.builder.getOrCreate()
```

Step 2

Once the connection is established, the underlying data can be read in. In this case, we are going to read in [data about Goodreads books from Kaggle](#).

```
df = spark_session.read.option("header", "true").csv("/content/drive/My
Drive/books.csv")
```

Step 3

Now that the data has been read into a data frame, the next step is to verify that the data has been successfully read in.

```
df.show()
```

Output Generated

bookID	title	authors	average_rating	isbn	isbn13	language_code	num_pages	ratings_count	text_reviews_count	publication_date	publisher
1	Harry Potter and ...	J.K. Rowling/Mary...	4.57	0439785960	9780439785969	eng	652	2095690	27591	9/16/2006	Scholastic Inc.
2	Harry Potter and ...	J.K. Rowling/Mary...	4.49	0439358078	9780439358071	eng	870	2153167	29221	9/1/2004	Scholastic Inc.
4	Harry Potter and ...	J.K. Rowling	4.42	0439554896	9780439554893	eng	352	6333	244	11/1/2003	Scholastic
5	Harry Potter and ...	J.K. Rowling/Mary...	4.56	043965548X	9780439655484	eng	435	2399585	36325	5/1/2004	Scholastic Inc.
8	Harry Potter Boxe...	J.K. Rowling/Mary...	4.78	0439682584	9780439682589	eng	2690	41428	164	9/13/2004	Scholastic
9	Unauthorized Harr...	W. Frederick Zimm...	3.74	0976540606	9780976540601	en-US	152	19	1	4/26/2005	Nimble Books
10	Harry Potter Coll...	J.K. Rowling	4.73	0439827604	9780439827607	eng	3342	28242	808	9/12/2005	Scholastic
12	The Ultimate Hitc...	Douglas Adams	4.38	0517226952	9780517226957	eng	815	3628	254	11/1/2005	Gramercy Books
13	The Ultimate Hitc...	Douglas Adams	4.38	0345453743	9780345453747	eng	815	249558	4080	4/30/2002	Del Rey Books
14	The Hitchhiker's ...	Douglas Adams	4.22	1400052920	9781400052929	eng	215	4930	460	8/3/2004	Crown
16	The Hitchhiker's ...	Douglas Adams/Ste...	4.22	0739322206	9780739322208	eng	6	1266	253	3/23/2005	Random House Audio
18	The Ultimate Hitc...	Douglas Adams	4.38	0517149257	9780517149256	eng	815	2877	195	1/17/1996	Wings Books
21	A Short History o...	Bill Bryson	4.21	076790818X	9780767908184	eng	544	248558	9396	9/14/2004	Broadway Books
22	Bill Bryson's Afr...	Bill Bryson	3.44	0767915062	9780767915069	eng	55	7270	499	12/3/2002	Broadway Books
23	Bryson's Dictiona...	Bill Bryson	3.87	0767910435	9780767910439	eng	256	2088	131	9/14/2004	Broadway Books
24	In a Sunburned Co...	Bill Bryson	4.07	0767903862	9780767903868	eng	335	72451	4245	5/15/2001	Broadway Books
25	I'm a Stranger He...	Bill Bryson	3.90	076790382X	9780767903820	eng	304	49240	2211	6/28/2000	Broadway Books
26	The Lost Continen...	Bill Bryson	3.83	0860920084	9780860920081	eng	299	45712	2257	8/28/1990	William Morrow Pa...
27	Neither Here nor ...	Bill Bryson	3.86	0380713802	9780380713806	eng	254	48701	2238	3/28/1993	William Morrow Pa...
28	Notes from a Sma...	Bill Bryson	3.91	0380727501	9780380727506	eng	324	80609	3301	5/28/1997	William Morrow Pa...

only showing top 28 rows

Figure 1: Example of output to verify data has been read in

Step 4

Register the DataFrame as an SQL temporary view so that you can interact with it using SQL commands.

```
df.createOrReplaceTempView("books")
```

Step 5

Query the temporary view using SQL.

```
sqlDF = spark_session.sql("SELECT * FROM books")
```

```
sqlDF.show()
```

Output Generated

bookID	title	authors	average_rating	isbn	isbn13	language_code	num_pages	ratings_count	text_reviews_count	publication_date	publisher
1	Harry Potter and ...	J.K. Rowling/Mary...	4.57	0439785960	9780439785969	eng	652	2095690	27591	9/16/2006	Scholastic Inc.
2	Harry Potter and ...	J.K. Rowling/Mary...	4.49	0439358078	9780439358071	eng	870	2153167	29221	9/1/2004	Scholastic Inc.
4	Harry Potter and ...	J.K. Rowling	4.42	0439554896	9780439554893	eng	352	6333	244	11/1/2003	Scholastic
5	Harry Potter and ...	J.K. Rowling/Mary...	4.56	043965548X	9780439655484	eng	435	2339585	36325	5/1/2004	Scholastic Inc.
8	Harry Potter Boxe...	J.K. Rowling/Mary...	4.78	0439682584	9780439682589	eng	2690	41428	164	9/13/2004	Scholastic
9	Unauthorized Harr...	W. Frederick Zimm...	3.74	0976540606	9780976540601	en-US	152	19	1	4/26/2005	Nimble Books
10	Harry Potter Coll...	J.K. Rowling	4.73	0439827604	9780439827607	eng	3342	28242	808	9/12/2005	Scholastic
12	The Ultimate Hitc...	Douglas Adams	4.38	0517226952	9780517226957	eng	815	3628	254	11/1/2005	Gramercy Books
13	The Ultimate Hitc...	Douglas Adams	4.38	0345453743	9780345453747	eng	815	249558	4080	4/30/2002	Del Rey Books
14	The Hitchhiker's ...	Douglas Adams	4.22	1400052920	9781400052929	eng	215	4930	460	8/3/2004	Crown
16	The Hitchhiker's ...	Douglas Adams/Ste...	4.22	0739322206	9780739322208	eng	6	1266	253	3/23/2005	Random House Audio
18	The Ultimate Hitc...	Douglas Adams	4.38	0517149257	9780517149256	eng	815	2877	195	1/17/1996	Wings Books
21	A Short History o...	Bill Bryson	4.21	076790818X	9780767908184	eng	544	248558	9396	9/14/2004	Broadway Books
22	Bill Bryson's Afr...	Bill Bryson	3.44	0767915062	9780767915069	eng	55	7270	499	12/3/2002	Broadway Books
23	Bryson's Dictiona...	Bill Bryson	3.87	0767910435	9780767910439	eng	256	2088	131	9/14/2004	Broadway Books
24	In a Sunburned Co...	Bill Bryson	4.07	0767903862	9780767903868	eng	335	72451	4245	5/15/2001	Broadway Books
25	I'm a Stranger He...	Bill Bryson	3.90	076790382X	9780767903820	eng	304	49240	2211	6/28/2000	Broadway Books
26	The Lost Continen...	Bill Bryson	3.83	0060920084	9780060920081	eng	299	45712	2257	8/28/1990	William Morrow Pa...
27	Neither Here nor ...	Bill Bryson	3.86	0380713802	9780380713806	eng	254	48701	2238	3/28/1993	William Morrow Pa...
28	Notes from a Smal...	Bill Bryson	3.91	0380727501	9780380727506	eng	324	80609	3301	5/28/1997	William Morrow Pa...

Figure 2: Example query of the temporary view using SQL

Step 6

Filter the data so it only records where the average rating is greater than 4.5.

```
sqlDF.filter("average_rating > 4.5").show()
```

Output Generated

bookID	title	authors	average_rating	isbn	isbn13	language_code	num_pages	ratings_count	text_reviews_count	publication_date	publisher
1	Harry Potter and ...	J.K. Rowling/Mary...	4.57	0439785960	9780439785969	eng	652	2095690	27591	9/16/2006	Scholastic Inc.
5	Harry Potter and ...	J.K. Rowling/Mary...	4.56	043965548X	9780439655484	eng	435	2339585	36325	5/1/2004	Scholastic Inc.
8	Harry Potter Boxe...	J.K. Rowling/Mary...	4.78	0439682584	9780439682589	eng	2690	41428	164	9/13/2004	Scholastic
10	Harry Potter Coll...	J.K. Rowling	4.73	0439827604	9780439827607	eng	3342	28242	808	9/12/2005	Scholastic
30	J.R.R. Tolkien 4...	J.R.R. Tolkien	4.59	0345538374	9780345538376	eng	1728	101233	1550	9/25/2012	Ballantine Books
36	The Lord of the R...	Chris Smith/Chr...	4.53	0618391002	9780618391004	eng	218	19822	46	11/5/2003	Houghton Mifflin ...
119	The Lord of the R...	Gary Russell	4.59	0618212906	9780618212903	eng	192	26153	102	6/12/2002	Houghton Mifflin ...
313	100 Years of Lync...	Ralph Ginzburg	4.61	0933121180	9780933121188	eng	270	88	4	11/22/1996	Black Classic Press
397	The Gettysburg Ad...	Abraham Lincoln/M...	4.53	0395883970	9780395883976	eng	32	5239	76	2/2/1998	HMH Books for You...
866	Fullmetal Alchemi...	Hiromu Arakawa/Ak...	4.57	142150460X	9781421504605	eng	192	9013	153	9/19/2006	VIZ Media LLC
868	Fullmetal Alchemi...	Hiromu Arakawa/Ak...	4.56	1591169259	9781591169253	eng	192	16666	299	9/13/2005	VIZ Media LLC
869	Fullmetal Alchemi...	Hiromu Arakawa/Ak...	4.57	1421504596	9781421504599	eng	192	11451	161	7/18/2006	VIZ Media LLC
871	Fullmetal Alchemi...	Hiromu Arakawa/Ak...	4.55	1591169291	9781591169291	eng	200	10752	294	11/8/2005	VIZ Media LLC
873	Fullmetal Alchemi...	Hiromu Arakawa/Ak...	4.52	1591169232	9781591169239	eng	192	14923	419	7/5/2005	VIZ Media LLC
955	The 5 Love Langua...	Gary Chapman	4.70	0802415318	9780802415318	eng	0	22	4	1/1/2005	Moody Publishers
1111	The Power Broker:...	Robert A. Caro	4.51	0394720245	9780394720241	eng	1344	11208	1237	7/12/1975	Vintage
1729	Metamorphoses: Vo...	Ovid/Frank Justus...	4.52	0674990471	9780674990470	eng	499	61	4	1/1/1985	Harvard Universit...
1750	Dionysiac Poetics...	Charles Segal	4.62	069101597X	9780691015972	eng	440	12	0	11/16/1997	Princeton Univers...
2005	Harry Potter and ...	J.K. Rowling	4.57	0747584664	9780747584667	eng	768	1213	78	6/23/2006	Bloomsbury Publis...
2034	Comediae 1: Acha...	Aristophanes/F.W...	5.00	0198145047	9780198145042	grc	364	0	0	2/22/1922	Oxford University...

only showing top 20 rows

Figure 3: Example of filtered data using SQL

Step 7

Filter the dataset to include books that have an average rating of greater than 4.5 and where the publisher is "Ballantine Books."

```
sqlDF.filter("average_rating > 4.5").filter("publisher == 'Ballantine Books '").show()
```

Output Generated

bookID	title	authors	average_rating	isbn	isbn13	language_code	num_pages	ratings_count	text_reviews_count	publication_date	publisher
30	J.R.R. Tolkien 4...	J.R.R. Tolkien	4.59	0345538374	9780345538376	eng	1728	101233	1550	9/25/2012	Ballantine Books

Figure 4: Example filtered dataset filtered by book average rating and specific publisher

Step 8

Add two new columns. For each book, depict what proportion of the overall ratings count are 'text review' counts. Cast the 'average_rating' string data type to decimal format.

```
import pyspark.sql.functions as F

sqlDF = sqlDF.withColumn("proportion_of_ratings",
F.round((sqlDF.text_reviews_count/sqlDF.ratings_count),2))

sqlDF = sqlDF.withColumn("avg_rating_decimal",
sqlDF.average_rating.cast('Decimal(4,2)'))

sqlDF.show()
```

Output Generated

e	authors	average_rating	isbn	isbn13	language_code	num_pages	ratings_count	text_reviews_count	publication_date	publisher	proportion_of_ratings	avg_rating_decimal
.	J.K. Rowling/Mary...	4.57	0439785960	9780439785969	eng	652	2095690	27591	9/16/2006	Scholastic Inc.	0.01	4.57
.	J.K. Rowling/Mary...	4.49	0439358078	9780439358071	eng	870	2153167	29221	9/1/2004	Scholastic Inc.	0.01	4.49
.	J.K. Rowling	4.42	0439554896	9780439554893	eng	352	6333	244	11/1/2003	Scholastic	0.04	4.42
.	J.K. Rowling/Mary...	4.56	043965548X	9780439655484	eng	435	2339585	36925	5/1/2004	Scholastic Inc.	0.02	4.56
.	J.K. Rowling/Mary...	4.78	0439682584	9780439682589	eng	2690	41428	164	9/13/2004	Scholastic	0.0	4.78
.	W. Frederick Zimm...	3.74	0976540606	9780976540601	en-US	152	19	1	4/26/2005	Nimble Books	0.05	3.74
.	J.K. Rowling	4.73	0439827604	9780439827607	eng	3342	28242	808	9/12/2005	Scholastic	0.03	4.73
.	Douglas Adams	4.38	0517226952	9780517226957	eng	815	3628	254	11/1/2005	Gramercy Books	0.07	4.38
.	Douglas Adams	4.38	0345453743	9780345453747	eng	815	249558	4080	4/30/2002	Del Rey Books	0.02	4.38
.	Douglas Adams	4.22	1408052920	9781408052929	eng	215	4930	460	8/3/2004	Crown	0.09	4.22
.	Douglas Adams/Ste...	4.22	0739322206	9780739322208	eng	6	1266	253	3/23/2005	Random House Audio	0.2	4.22
.	Douglas Adams	4.38	0517149257	9780517149256	eng	815	2877	195	1/17/1996	Wings Books	0.07	4.38
.	Bill Bryson	4.21	076790818X	9780767908184	eng	544	248558	9396	9/14/2004	Broadway Books	0.04	4.21
.	Bill Bryson	3.44	0767915062	9780767915069	eng	55	7270	499	12/3/2002	Broadway Books	0.07	3.44
.	Bill Bryson	3.87	0767910435	9780767910439	eng	256	2088	131	9/14/2004	Broadway Books	0.06	3.87
.	Bill Bryson	4.07	0767903862	9780767903868	eng	335	72451	4245	5/15/2001	Broadway Books	0.06	4.07
.	Bill Bryson	3.90	076790382X	9780767903820	eng	304	49240	2211	6/28/2000	Broadway Books	0.04	3.90
.	Bill Bryson	3.83	0060920004	9780060920001	eng	299	45712	2257	8/28/1990	William Morrow Pa...	0.05	3.83
.	Bill Bryson	3.86	0380713002	9780380713006	eng	254	48701	2238	3/28/1993	William Morrow Pa...	0.05	3.86
.	Bill Bryson	3.91	0380727501	9780380727506	eng	324	80609	3301	5/28/1997	William Morrow Pa...	0.04	3.91

Figure 5: Example dataset output depicting the proportion of text review counts in decimal format

Step 9

(NOTE: From this step onwards, we will begin aggregating data using *GroupBy*.)

Find the lowest average_rating where the publisher is "Broadway Books."

```
sqlDF.filter("publisher == 'Broadway  
Books']").groupBy().min("avg_rating_decimal").show()
```

Output Generated

```
+-----+  
|min(avg_rating_decimal)|  
+-----+  
|                      3.25|  
+-----+
```

Figure 6: Example aggregate data of lowest average rating

Step 10

Find the highest average_rating where the publisher is "Broadway Books."

```
sqlDF.filter("publisher == 'Broadway  
Books']").groupBy().max("avg_rating_decimal").show()
```

Output Generated

```
+-----+  
|max(avg_rating_decimal)|  
+-----+  
|                      4.43|  
+-----+
```

Figure 7: Example aggregate data of highest average rating

Step 11

Ascertain the cumulative count of ratings received for "Broadway Books."

#First convert from string type to integer

```
sqlDF = sqlDF.withColumn("total_rating_count",
```

```
sqlDF.ratings_count.cast('Integer'))

sqlDF.filter("publisher == 'Broadway
Books']").groupBy().sum("total_rating_count").show()
```

Output Generated

```
+-----+
|sum(total_rating_count)|
+-----+
|                695164|
+-----+
```

Figure 8: Example aggregate data of cumulative ratings

Step 12

Ascertain the average count of ratings received for “Broadway Books.”

```
sqlDF.filter("publisher == 'Broadway
Books']").groupBy().avg("total_rating_count").show()
```

Output Generated

```
+-----+
|avg(total_rating_count)|
+-----+
|    14187.020408163266|
+-----+
```

Figure 9: Example data for average count of ratings

Step 13

Show average count of ratings for each publisher.

```
sqlDF.groupBy("publisher").avg("total_rating_count").show()
```

```

+-----+-----+
| publisher | avg(total_rating_count) |
+-----+-----+
| The New Press | 1693.0 |
| Chosen Books | 536.5 |
| Digireads.com | 24628.16666666666668 |
| Ember | 351258.4 |
| IVP Books | 10.5 |
| No Exit Press | 43253.8 |
| Dabel Brothers Pr... | 276.0 |
| DAW | 5079.1463414634145 |
| Celestial Arts | 442.0 |
| Arcadia Publishin... | 5.0 |
| Hachette Littérature | 1.0 |
| Cleis Press | 428.5 |
| Chicago Review Press | 6315.857142857143 |
| HQN | 7769.0 |
| World Wrestling E... | 1154.0 |
| Wayne State Unive... | 45.0 |
| Doubleday | 8012.916666666667 |
| Verso | 1204.8666666666666 |
| Time Life Medical | 60.2 |
| Roc | 16373.5 |
+-----+-----+
only showing top 20 rows

```

Figure 10: Example data for average count of ratings for each publisher

Step 14

Compute the standard deviation of ratings count by publisher.

```

sqlDF.groupBy("publisher").avg("total_rating_count").show()

# Standard deviation of ratings count by publisher

sqlDF.groupBy("publisher").agg(F.stddev("ratings_count")).show()

```

Output Generated

```
+-----+-----+
| publisher | stddev_samp(ratings_count) |
+-----+-----+
| The New Press | 4021.5151825694215 |
| Chosen Books | 457.49808742769625 |
| Digireads.com | 51201.525162505335 |
| Ember | 691000.7816122931 |
| IVP Books | 14.849242404917497 |
| No Exit Press | 17332.11097068098 |
| Dabel Brothers Pr... | NaN |
| DAW | 6372.894532945668 |
| Celestial Arts | 612.3544725075502 |
| Arcadia Publishin... | 2.9439202887759492 |
| Hachette Littérature | NaN |
| Cleis Press | 488.69792987761537 |
| Chicago Review Press | 9251.830511283904 |
| HQN | 7846.056844045931 |
| World Wrestling E... | NaN |
| Wayne State Unive... | 46.66904755831214 |
| Doubleday | 24104.695025350382 |
| Verso | 1821.766994143977 |
| Time Life Medical | 40.95200442794793 |
| Roc | 31089.472770422744 |
+-----+-----+
only showing top 20 rows
```

Figure 11: Example data for standard deviation of ratings count by publisher

Joining Datasets using Spark SQL

Below is a list of steps to join datasets using Spark SQL.

Step 1

Create the tables that will be joined.

```
# Create the tables that will be joined
```

```
valuesA = [('Dog',1),('Monkey',2),('Elephant',3),('Penguin',4)]
```

```
TableA = spark_session.createDataFrame(valuesA,['name','id'])
```

```
valuesB = [('Lizard',1),('Dog',2),('Elephant',3),('Rat',4)]
```

```
TableB = spark_session.createDataFrame(valuesB,['name','id'])
```

```
TableA.show()
```

```
TableB.show()
```

Step 2

Specify aliases for the two tables.

```
# Create table aliases
```

```
T1 = TableA.alias('T1')
```

```
T2 = TableB.alias('T2')
```

Step 3

Inner join on name (Example): It returns matching records from the two tables, respectively.

```
T1.join(T2, T1.name == T2.name, 'inner').join(T2, T2.name == T2.name).show()
```

Output Generated

name	id	name	id
Elephant	3	Elephant	3
Dog	1	Dog	2

Figure 12: Example data for inner join on name

Step 4

Left Join (Example): Choose "left" to select all records from table 1 and return data from table two when matches occur.

```
T1.join(T2, T1.name == T2.name, 'left').show()
```

Output Generated

name	id	name	id
Elephant	3	Elephant	3
Monkey	2	null	null
Dog	1	Dog	2
Penguin	4	null	null

Figure 13: Example data for left join

Step 5

Filter out the nulls.

```
left_join = T1.join(T2, T1.name == T2.name, 'left')
```

```
left_join.filter((T2.name).isNull()).show()
```

Output Generated

name	id	name	id
Elephant	3	Elephant	3
Dog	1	Dog	2

Figure 14: Example data after filtering out the nulls

Step 6

Right Join (Example): Choose “right” to select all records from table 2 and return data from table 1 when matches occur.

```
T1.join(T2, T1.name == T2.name, 'right').show()
```

Output Generated

name	id	name	id
Elephant	3	Elephant	3
null	null	Rat	4
null	null	Lizard	1
Dog	1	Dog	2

Figure 15: Example data for right join

Step 7

Full Outer Join (Example): The output from a full outer join returns all records from both tables with nulls where the two do not match.

```
T1.join(T2, T1.name == T2.name, how='full').show()
```

Output Generated

name	id	name	id
Elephant	3	Elephant	3
null	null	Rat	4
null	null	Lizard	1
Monkey	2	null	null
Dog	1	Dog	2
Penguin	4	null	null

Figure 16: Example data for full outer join

References

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